Answer Key & Teacher Notes

Session 1 — Drawing Nets of Solids · Session 2 — Reading Nets and Solids

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Drawing Nets of Solids

TALK PROMPTS

- How do a solid's attributes tell you what its net has to look like?
- How can you tell from your net which rectangles fold onto each other?
- Why are all four triangle faces of the paper pyramid the same size?

WATCH FOR

- Leaving out a face (or drawing one twice) so the net does not close.
- Joining edges of different lengths, so the faces do not meet when folded.
- Thinking any 6 squares in a row fold into a cube.

PARTNER TASK

Cereal Box. Partners unfold a real or paper box (12 in by 8 in by 2 in), trace the net, label every face, and color matching pairs the same color. Then find the area of the largest face (96 in²) and compare nets: are there different correct nets?

SESSION 2 · TEACHER NOTES

Reading Nets and Solids

TALK PROMPTS

- How can a net tell you the name of a solid before anyone folds it?
- How does a square base change this net from an ordinary box net?

WATCH FOR

- Counting edges and vertices on the flat net instead of the folded solid.
- Calling any net with triangles a pyramid (a triangular prism has triangles too).
- Naming a solid by the number of shapes instead of by its base shape.

PARTNER TASK

Prism or Pyramid. Partner A sketches a net with 2 hexagons and 6 rectangles; Partner B sketches a net with 1 hexagon and 6 triangles. Each names the other's solid and counts its faces (8 and 7), then checks by folding paper copies.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.4	1	1	1	1	1	1	3	2	2	1	1
S1 · Form B	6.G.A.4	1	1	1	1	1	1	3	2	2	1	1
S2 · Form A	6.G.A.4	1	2	2	2	1	2	2	3	2	1	1
S2 · Form B	6.G.A.4	1	2	2	2	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

net: A flat **pattern** that folds into a solid.

face: One **flat** side of a solid.

prism: A solid with two matching **bases** joined by rectangles.

pyramid: A solid with one base and **triangle** faces that meet at a point.

- Every face of the solid appears **once** in its net.
- Edges that fold together must have the same **length**.
- A rectangular prism net has **6** rectangles in 3 matching pairs.

Example: Bottom and top: two **5** cm by 3 cm rectangles. Front and back: two 5 cm by **2** cm rectangles. Left and right: two **3** cm by 2 cm rectangles. Answer: **6** rectangles in all

YOU TRY

1. a. $4 \times 4 = 16$ cm² b. $\frac{1}{2} \times 4 \times 3 = 6$ cm²

2. a. 6 b. 5

EXIT TICKET

1. a. 6 b. 4

2. $\frac{1}{2} \times 10 \times 8 = 40$ cm²

3. **B** (2 triangles + 3 rectangles)

PRACTICE FORM A

1. a. 6 b. 5 c. 2 d. 6

2. a. 5 (1 square + 4 triangles) b. $6 \times 6 = 36$ in² c. $\frac{1}{2} \times 6 \times 5 = 15$ in²

3. a. 6 rectangles b. 96 in² (the two largest edges multiplied)

4. a. 2 triangles (and 3 rectangles) b. $\frac{1}{2} \times 7 \times 4 = 14$ ft²

5. 6 faces; $9 \times 9 = 81$ cm²

6. $\frac{1}{2} \times 14 \times 11 = 77$ in²

7. No. A cube has 6 faces, so its net needs exactly 6 squares. With 5 squares, one face would be missing.

8. a. two 10×5, two 10×4, two 5×4 b. $10 \times 5 = 50$ in²

9. A square has 4 sides, so a square pyramid has 4 triangle faces. The net needs one triangle on each of the 4 sides of the square; with only 3, one face is missing.

10. **A** (solid)

11. **B** (1 square and 4 triangles)

Self-check word **UNFOLDS**

PRACTICE FORM B

1. a. 6 b. 5 c. 3 d. 1

2. a. 5 (1 square + 4 triangles) b. $8 \times 8 = 64$ in² c. $\frac{1}{2} \times 8 \times 7 = 28$ in²

3. a. 6 rectangles b. 90 in² (the two largest edges multiplied)

4. a. 2 triangles (and 3 rectangles) b. $\frac{1}{2} \times 6 \times 5 = 15$ ft²

5. 6 faces; $7 \times 7 = 49$ cm²

6. $\frac{1}{2} \times 10 \times 7 = 35$ in²

7. No. A cube has 6 faces, so its net needs exactly 6 squares. With 7 squares, one square would overlap another face.

8. a. two 10×7, two 10×3, two 7×3 b. $10 \times 7 = 70$ in²

9. A triangular prism has 5 faces: 2 triangle bases and 3 rectangles (one for each side of the triangle). The net is missing one rectangle.

10. **C** (faces)

11. **C** (6)

Self-check word **FOLDING**

GUIDED NOTES

edge: A **segment** where two faces meet.

vertex: A **corner** where edges meet.

base: The face that **names** a prism or a pyramid.

- A prism has **two** matching bases joined by rectangles.
- A pyramid has **one** base and triangles that meet at a point.
- Count edges and vertices on the **folded** solid, not on the flat net.

Example: Pieces: 1 + 4 = **5** faces. One square base with triangles: a square **pyramid**. Edges: 4 + 4 = **8**. Vertices: 4 + 1 = 5. Answer: square pyramid: 5 faces, 8 edges, **5** vertices

YOU TRY

1. a. triangular prism b. 5 faces

2. a. octagonal prism b. hexagonal pyramid

EXIT TICKET

1. a. pentagonal pyramid b. triangular prism

2. Base 4 × 4 = 16; triangles 4 × (½ × 4 × 6) = 48; total 64 in²

3. **B** (8)

PRACTICE FORM A

1. a. pentagonal prism b. triangular pyramid c. cube d. octagonal prism

2. a. cube b. 6 faces, 12 edges, 8 vertices c. 4 × 4 = 16 cm² d. 6 × 16 = 96 cm²

3. a. rectangular prism, 9 in × 5 in × 3 in b. 2 × (45 + 27 + 15) = 174 in²

4. a. square (rectangular) prism, 12 in × 12 in × 3 in b. 2 × 144 + 4 × 36 = 432 in²

5. a. 6 faces, 12 edges, 8 vertices b. 7 × 7 = 49 cm²

6. No. One base with triangles that meet at a point is a pyramid: a square pyramid.

7. Triangular pyramid; one outer triangle ½ × 9 × 8 = 36 cm²

8. a. Net 1 triangular prism; Net 2 square pyramid; Net 3 rectangular prism b. Two matching bases joined by rectangles make a prism; one base with triangles meeting at a point makes a pyramid. The base shape gives the name.

9. Count on the folded solid. A cube has 6 faces, 12 edges, and 8 vertices.

10. **A** (bases)

11. **B** (hexagonal pyramid)

Self-check word **SOLIDS**

PRACTICE FORM B

1. a. hexagonal prism b. pentagonal pyramid c. triangular prism d. rectangular pyramid

2. a. cube b. 6 faces, 12 edges, 8 vertices c. 5 × 5 = 25 cm² d. 6 × 25 = 150 cm²

3. a. rectangular prism, 10 in × 6 in × 4 in b. 2 × (60 + 40 + 24) = 248 in²

4. a. square (rectangular) prism, 10 in × 10 in × 2 in b. 2 × 100 + 4 × 20 = 280 in²

5. a. 6 faces, 12 edges, 8 vertices b. 9 × 9 = 81 cm²

6. No. One base with triangles that meet at a point is a pyramid: a pentagonal pyramid.

7. Triangular pyramid; one outer triangle ½ × 10 × 6 = 30 cm²

8. a. Net 1 hexagonal prism; Net 2 triangular pyramid; Net 3 cube b. Two matching bases joined by rectangles make a prism; one base with triangles meeting at a point makes a pyramid. The base shape gives the name.

9. Count on the folded solid. A square pyramid has 5 faces (1 square + 4 triangles), 8 edges, and 5 vertices.

10. **B** (1)

11. **C** (cube)

Self-check word **SHAPES**

SHEET 1 · DRAWING NETS OF SOLIDS

1. Net of 6 rectangles: two 6 in × 2 in, two 6 in × 4 in, two 2 in × 4 in, joined so they fold with no overlaps.
2. a. 6 b. 4 c. 3 d. 5
3. a. $8 \times 8 = 64$ cm² b. $\frac{1}{2} \times 8 \times 6 = 24$ cm² All 4 triangles: 96 cm²
4. two 8 in × 3 in, two 8 in × 12 in, two 3 in × 12 in
5. No. The row wraps around 4 sides, and the last 2 squares overlap the first 2. Nothing covers the top or bottom. A cube net needs squares attached above and below the row.

SHEET 2 · READING NETS AND SOLIDS

1. triangular prism; 5 faces
2. a. rectangular pyramid b. hexagonal prism c. triangular prism d. pentagonal pyramid e. cube f. pentagonal prism
3. a. $6 \times 6 = 36$ in² b. $4 \times (\frac{1}{2} \times 6 \times 5) = 60$ in²
4. Rectangular prism (square prism), 4 cm × 4 cm × 9 cm
5. **C** (9)
6. **B** (7)

October 2026 conceptual additions — teacher key

Session 1 · Drawing Nets of Solids

Two 4×3 faces, two 4×2 faces, and two 3×2 faces; shared edges must have matching lengths. A valid arrangement must fold without overlap, not merely contain these six faces.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Two 4×3 faces, two 4×2 faces, and two 3×2 faces; shared edges must have matching lengths. A valid arrangement must fold without overlap, not merely contain these six faces.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Reading Nets and Solids

The cross-like second arrangement is a cube net; the row of six folds with overlaps and cannot close as a cube. Face count alone is insufficient.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: The cross-like second arrangement is a cube net; the row of six folds with overlaps and cannot close as a cube. Face count alone is insufficient.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.